

Network Theory – BAECE103

Dr. Sathya P
Associate Professor
SENSE

Faculty Details

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Course Objectives

1. Familiarize with various circuit laws and theorems to solve DC and AC circuits.
2. Understand advanced mathematical methods such as the Laplace transform and graph theory for solving circuit problems.
3. Develop the ability to analyze the two-port networks, first and second-order circuits.

Course Outcomes

1. Compute DC circuit parameters using circuit laws and theorems.
2. Determine the natural and step responses of RC, RL, and RLC circuits.
3. Apply appropriate techniques and theorems for sinusoidal steady-state analysis.
4. Apply the Laplace transform to circuits for steady-state and transient analysis.
5. Compute two-port network parameters and network graph matrices for circuit analysis.

Syllabus

Module 1 : DC circuit Analysis and Theorems - (12 hours)

- Basic Circuit Elements and Sources
- Ohm's Law and Kirchhoff's Laws;
- Series Resistors and Voltage Division
- Parallel Resistors and Current Division
- Wye-Delta Transformations;
- Methods of Analysis: Nodal Analysis, Mesh Analysis
- **Theorems:**
 - Superposition theorem
 - Source Transformation Theorem
 - Thevenin's theorem
 - Norton's theorem
 - Maximum power transfer theorem
 - Reciprocity theorem

Module 2 : Transient Analysis of First and Second order circuits - (12 hours)

- First order circuits
 - Source Free RC Circuit
 - Source Free RL Circuit
- Singularity Functions
- Step Response of an RC Circuit
- Step Response of an RL Circuit
- Second order circuits
 - Source Free Series RLC Circuit
 - Source Free Parallel RLC Circuit
 - Step Response of a Series RLC Circuit
 - Step Response of a Parallel RLC Circuit
- General Second order circuits

Module 3 : Sinusoidal Steady State Analysis - (12 hours)

- Sinusoids, Phasors, Phasor relationships for circuit elements, Impedance and Admittance
- Kirchhoff's Laws in the frequency domain
- Instantaneous and Average power, complex power
- Methods of analysis: Node voltage and mesh current analysis
- **Theorems:**
 - Superposition theorem
 - Source Transformation theorem
 - Thevenin's theorem
 - Norton's theorem
 - Maximum power transfer theorem

Module 4 : Circuit Analysis in the S domain – (12 hours)

- Circuit Element models
- Circuit Analysis with and without initial conditions
- Transfer functions
- Laplace to Fourier transform
- Frequency domain analysis of Linear circuits
- **Passive Filters**
 - Low Pass Filter
 - High Pass Filter
 - Band Pass Filter
 - Band Stop Filter

Module 5 : Two port Networks and Network Graphs – (10 hours)

- Two Port Networks:
 - Impedance Parameters
 - Admittance Parameters
 - Hybrid Parameters
 - Transmission Parameters
- Relationships between Parameters
- Interconnection of Networks
- Network graphs:
 - Incidence matrix
 - Cut set matrix
 - Tie set matrix

Module 6 : Contemporary Issues – (2 hours)

- Guest lecture by an Industry expert
- Industrial visit

Text Book and Reference Books

- Charles K. Alexander, Matthew N. O. Sadiku, “ Fundamentals of Electric Circuits”, 2020, 7th Edition, McGraw Hill Higher Education, United States.
- Abhijit Chakrabarti, “Circuit Theory Analysis and Synthesis”, 2018, 7th Revised Edition, Dhanpat Rai & Co., India.
- A. Sudhakar, Shyammohan S. Palli, “Circuits and Networks: Analysis and Synthesis”, 2017, 5th Edition, McGraw Hill, India.
- M.E. Van Valkenburg, T.S. Rathore, “Network Analysis”, 2019, Revised 3rd Edition, Pearson Education, England.
- Mahmood Nahvi, Joseph A. Edminister, “Schaum’s Outline of Electric Circuits”, 2018, 7th Edition, McGraw-Hill Education, United States.
- W.H. Hayt, J.E. Kemmerly & S.M. Durbin, “ Engineering Circuit Analysis”, 2019, 9th edition, McGraw Hill Higher Education, United States.

Evaluation Methodology

- Continuous Assessment Test I (CAT-I) – 15 Marks : Portion- Module 1 & 2
- Continuous Assessment Test II (CAT-II) – 15 Marks : Portion – Module 3 & 4.
- Quiz 1 – 10 Marks : Portion – Module 1 & 2
- Course Project – 20 Marks : Due date- 04.11.2025
- Final Assessment test – 100 Marks
- Continuous Assessment Marks – 60 Marks (60 %)
- FAT – 40 %
- Total – 100 %